

VITORIA AP, Spain

WMO: 080800

Lat: 42.872N

Lon: 2.733W

Elev: 1682

StdP: 13.82

Time Zone: 1.00 (EUC)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.6	27.4	21.3	17.0	28.6	24.7	19.9	31.1	27.4	51.4	23.6	49.5	1.9	20	0.518

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	25.6	92.2	68.3	87.7	67.4	83.8	66.4	70.9	85.9	69.2	83.5	67.5	80.3	7.7	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.4	103.4	75.8	64.6	97.2	73.5	63.1	91.8	71.5	35.7	86.0	34.2	83.1	32.8	80.6	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.2	18.1	15.8	DB	17.9	99.6	4.8	3.3	14.4	101.9	11.6	103.8	8.9	105.7	5.5	108.1
			WB	17.5	74.7	4.9	1.9	14.0	76.1	11.2	77.2	8.4	78.3	4.9	79.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.1	42.0	42.6	47.6	50.9	56.9	63.7	67.3	68.0	63.0	56.7	47.3	42.3	(6)
	DBStd	11.20	5.86	5.89	6.11	5.81	6.32	6.65	5.70	6.02	5.71	6.16	5.82	6.06	(7)
	HDD50	1034	255	214	117	57	12	0	0	0	0	15	118	246	(8)
	HDD65	4341	713	627	538	422	262	104	40	35	104	263	530	703	(9)
	CDD50	2536	7	7	44	85	226	411	538	559	390	223	38	8	(10)
	CDD65	367	0	0	0	0	10	65	113	129	43	7	0	0	(11)
	CDH74	4404	0	0	5	38	252	868	1198	1385	573	85	0	0	(12)
	CDH80	1738	0	0	0	3	66	350	489	636	186	8	0	0	(13)
(14) Wind	WSAvg	5.7	6.1	6.2	6.3	6.1	5.7	5.7	6.0	5.5	4.8	4.9	5.7	5.8	(14)
(15) Precipitation	PrecAvg	39.2	4.2	4.0	3.9	3.7	3.2	2.3	1.4	1.4	2.3	3.4	5.4	4.1	(15)
	PrecMax	59.0	10.9	11.2	9.2	7.9	8.7	5.8	3.9	3.3	4.7	12.8	10.3	7.9	(16)
	PrecMin	30.1	0.2	0.5	0.6	1.0	1.0	0.6	0.1	0.4	0.9	0.4	1.7	0.6	(17)
	PrecStd	6.4	2.5	2.8	2.3	1.7	1.6	1.3	0.9	0.8	1.1	2.4	2.0	2.0	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.1	65.9	73.5	78.9	86.9	94.9	96.1	98.4	91.5	80.7	66.6	60.3	(19)
		MCWB	53.2	51.8	56.0	60.0	64.9	69.2	69.1	69.3	67.0	62.3	57.1	53.3	(20)
	2%	DB	56.7	59.9	68.3	73.1	80.5	88.3	90.4	92.8	85.1	76.3	62.8	56.9	(21)
		MCWB	50.9	50.3	53.9	57.4	63.2	67.5	68.2	68.4	65.0	61.2	54.8	51.2	(22)
	5%	DB	53.7	56.1	64.3	68.0	75.5	83.9	85.7	87.4	80.6	71.8	59.4	53.9	(23)
		MCWB	48.7	48.8	52.4	55.6	60.8	66.5	67.4	67.8	64.4	59.8	53.2	49.0	(24)
	10%	DB	51.4	53.2	59.9	63.2	70.9	78.5	80.7	82.0	76.4	68.2	56.9	51.7	(25)
		MCWB	47.0	47.3	50.7	53.1	59.0	64.4	66.8	66.8	63.0	58.8	51.7	47.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.9	53.8	57.8	61.5	67.3	71.7	73.1	73.3	69.7	65.2	59.4	53.9	(27)
		MCDB	58.5	61.3	69.3	76.0	82.2	87.9	86.2	89.5	84.9	75.8	64.2	58.6	(28)
	2%	WB	51.4	51.3	55.1	58.5	64.2	69.3	70.8	70.9	67.3	62.9	56.0	51.5	(29)
		MCDB	55.3	58.2	65.3	70.5	78.2	84.6	84.7	86.4	80.5	72.3	61.3	55.6	(30)
	5%	WB	49.3	49.5	53.1	56.2	61.9	67.3	69.0	69.0	65.6	61.2	54.1	49.9	(31)
		MCDB	53.0	55.2	62.1	66.1	73.6	81.0	82.1	83.3	76.9	69.0	58.6	53.4	(32)
	10%	WB	47.5	47.7	51.1	54.0	59.7	65.2	67.1	67.2	63.8	59.6	52.2	48.2	(33)
		MCDB	50.5	52.2	59.0	62.1	69.3	76.9	78.5	79.2	73.5	66.6	56.1	51.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	14.0	16.6	20.6	21.2	22.7	24.2	24.6	25.6	24.8	21.1	14.8	13.9	(35)
		MCDBR	17.6	24.7	31.4	32.5	33.6	35.3	35.1	36.6	33.9	28.3	20.1	16.5	(36)
	5% WB	MCWBR	12.2	15.6	17.8	18.4	17.6	16.5	15.7	15.1	15.7	14.8	13.1	11.8	(37)
		MCWBR	15.5	22.3	27.2	28.6	30.9	31.7	29.7	32.2	30.2	23.8	17.7	14.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.302	0.316	0.349	0.369	0.382	0.392	0.383	0.383	0.364	0.340	0.316	0.299	(40)
		taud	2.528	2.480	2.398	2.349	2.339	2.341	2.364	2.373	2.427	2.500	2.526	2.539	(41)
	Ebn at Noon		268	282	283	285	283	279	280	277	275	269	259	259	(42)
		Edh at Noon	22	27	33	38	39	39	38	36	32	26	22	20	(43)
(44) All-Sky Solar Radiation	RadAvg		478	731	1124	1466	1756	1981	2043	1787	1400	913	543	439	(44)
	RadStd		54	98	144	152	137	125	99	95	68	81	60	53	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page